



Becoming someone in Esports: Identity, aspiration, and psychological commitment in a digital entertainment ecosystem

Arianna Boldi ^{*} , Domenico Sanseverino

Department of Psychology, University of Turin, Torino, Italy

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ABSTRACT

The classification of esports participants remains conceptually fragmented and empirically inconsistent, particularly in informal ecosystems lacking standardized professional criteria. This study examines how individuals involved in the Italian esports scene define their roles, occupational identities, and career aspirations within the precarious and loosely regulated esports market. Drawing on survey data from 297 gamers, including casual gamers and esports participants, we explore differences in gaming practices, digital skills, employment status, and career orientation. Informed by theories of aspirational labor and boundaryless careers, we propose a role-based taxonomy that recognizes functional roles such as administrators, content creators, and hybrid professionals. Esports participants differ from casual gamers in gaming intensity, platform preferences, and digital collaboration skills, but not in employment status or career values. Crucially, perceived professional commitment is more closely tied to functional roles than to gaming frequency. Most esports participants do not consider their engagement economically viable, underscoring the aspirational and precarious nature of esports labor. By situating player self-perceptions within broader dynamics of gaming platforms and digital entertainment, this study advances a context-sensitive model of esports engagement and contributes to comparative research and policy debates on esports and digital entertainment.

1. Introduction

Esports have become a global phenomenon, attracting millions of players across platforms and games [48]. In countries like South Korea, China, and the United States, the field has undergone increasing institutionalization [45]. In contrast, the Italian esports ecosystem remains largely informal and under-regulated, with few career pathways, weak institutional support, and limited legal recognition [40]. As a result, the professional status of esports participants remains ambiguous. This lack of institutionalization hinders both industry development and academic theorization. Although recent studies have attempted to define “esports athletes” [8], most rely on fragmented criteria, such as tournament participation [51], contractual affiliation [28], or performance ranking [27,36]. Such inconsistencies limit cross-study comparability and obstruct a coherent understanding of esports as a professional domain. Moreover, much of the literature overlooks the broader ecology of esports work. Attention tends to focus on players, neglecting roles such as streamers, team managers, or content creators. This omission is particularly problematic in informal contexts like Italy, where individuals often engage in overlapping roles without stable occupational

identity [7]. While these actors contribute to the digital economy, we still lack systematic knowledge of their competencies, values, and career orientation. First, little is known about the digital competences mobilized by esports participants—such as communication, collaboration, strategic thinking, and content creation [54]. These are key to navigating both the esports field and broader digital labor markets. Second, research has not addressed how esports actors experience job insecurity or interpret their engagement in terms of professional aspirations. Constructs like job insecurity [47] and career values [12] are well established in organizational psychology but rarely applied to esports or platform labor. As such, key questions remain: Do participants see their involvement as a hobby, a side job, or a potential career? What values guide their commitment? And how do they assess the sustainability of their trajectories? Understanding this population is crucial. According to different reports, in 2023, an estimated 13 million Italians played video games regularly; of these, 7.3 million followed esports content, and 3.3 million watched competitions weekly [9,25]. Far from a marginal subculture, esports represent an emerging domain of digital labor, blurring the boundaries between leisure and work [10]. This study addresses two main questions: **RQ1: In what ways do esports participants differ from**

^{*} Corresponding author at: University of Turin, Department of Psychology, Via Verdi 10, 10124 Torino, Italy.

E-mail address: arianna.boldi@unito.it (A. Boldi).

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casual gamers in gaming practices, digital skills, career values, and perceptions of job insecurity? **RQ2:** Among esports participants, how are functional roles (e.g., player, content creator, administrator) and levels of commitment related to professional identity and perceptions of career sustainability?

We address these questions via a survey of 297 Italian gamers. To operationalize esports labor, we used: (a) the DigComp 2.2 framework [54] to assess the self-perception of digital competence; (b) Schein's career anchors (1990) to explore value-based orientations; and (c) a multidimensional measure of job insecurity [35]. Our results are interpreted within the frameworks of aspirational labor [17] and boundaryless careers [4] and culminate in a role-based taxonomy of esports engagement, offering a heuristic model to analyze career commitment in structurally informal labor contexts.

2. Literature review

This section provides a structured theoretical foundation that directly addresses the two central research questions of our study: (RQ1) identifying the differences between esports participants and casual gamers, particularly regarding the self-perception of digital competence, career values, and job insecurity perceptions, and (RQ2) exploring how functional roles and perceived professional commitment differentiate participants within the esports ecosystem Table 1 summarizes current gaps in the literature.

2.1. Differences between esports participants and casual gamers

While the majority of existing studies define esports actors primarily as “players,” this study adopts a broader operational category: esports participants. This term is used to capture not only competitive players, but also individuals who contribute to the esports ecosystem in non-playing roles, including content creators, coaches, administrators, and hybrid figures. Our rationale aligns with recent calls for expanding the analytical focus beyond gameplay alone, particularly in informal markets such as Italy where occupational boundaries are fluid and roles are frequently hybridized [7]. By adopting “participant” as an inclusive category, we aim to more accurately reflect the heterogeneity of esports engagement and its socio-professional implications—an aspect largely overlooked in existing player-centric literature.

2.1.1. Conceptualizing esports players and casual gamers

In academic literature, esports players are commonly defined as individuals who engage in competitive gaming activities with a structured and goal-oriented approach. For instance, Trotter et al. [51] describe esports athletes as those who participate in esports competitions and possess an official ranking within their respective games. Similarly, Khromov et al. [28] distinguish between esports athletes, who have professional contracts with esports teams, and players, who, while lacking formal contracts, exhibit significant skill levels and competitive engagement. In contrast, casual gamers are characterized by their

engagement with video games primarily for entertainment and relaxation purposes. Their gaming behavior is typically flexible, episodic, and low-stakes, without a focus on structured progression or professional aspirations [24]. However, clearly distinguishing between esports players and casual gamers remains conceptually challenging. Pedraza-Ramirez et al. [36] highlight the ambiguity in defining an esports athlete, noting that high-ranking competitive play can coexist with informal or amateur status. Formosa et al. [19] propose a spectrum model of engagement but acknowledge the lack of operational criteria for distinguishing between recreational and professionalized forms of participation. Bubna et al. [8] emphasize the relevance of self-identification, suggesting that esports identity is not reducible to contractual or performance-based metrics. These divergences complicate comparative research and raise important questions about how to define the boundaries of esports involvement.

2.1.2. Digital competence and skill Sets

According to the European DigComp framework, digital competence is defined as the confident, critical, and responsible use of digital technologies for learning, work, and participation in society, and comprises five areas: information and data literacy, communication and collaboration, digital content creation, safety, and problem-solving [54]. The conceptual foundation of the framework highlights general ICT literacy as a dimension of the broader dimension of digital literacy, which sits at the intersection of information and media literacy; as such, the framework does not equate digital literacy with digital competence, the latter of which is intended as the combination of knowledge, skills, and attitudes, including both digital and non-digital aspects [3]. In the context of gaming, several studies have documented how digital gameplay, especially in esports, cultivates competencies such as teamwork, communication, and real-time decision-making[59]. These are particularly evident in multiplayer and competitive environments, where strategic coordination and role specialization are critical. Research by Ball et al. [6] suggests that gameplay can enhance computer self-efficacy and ICT-related skills, especially in digitally marginalized populations.

Complementing this competence perspective is research on digital self-efficacy (DSE) which clarifies the role of confidence in executing digital tasks, namely “the belief in one’s capabilities to organize and execute courses of Internet actions required to produce given attainments” [18]. Self-efficacy, considered as the mental representation of one’s abilities, is important because it can significantly affect the development of digital competences [38]. Within technology-adoption models, for instance, perceived ease of use (PEOU), that is the degree to which a person believes that using a particular system would be free of effort, and perceived usefulness jointly predict intention and use [15], while computer self-efficacy affects intention indirectly, by shaping effort expectancy and PEOU [53]. Recent work in other domains (consumer contexts, [2]) cautions against assuming uniformly positive effects of DSE on the adoption of technology and reinforces our distinction between perception of competence (focus of this study) and confidence in enactment.

Since esports participants are often required to develop content for social media, manage digital identities, and perform in real-time mediated environments—tasks that casual gamers are not typically expected to perform—there might be differences between the two categories. Musa [34] emphasizes that casual gaming environments can support a complex set of digital literacies, such as cognitive, socio-emotional, and procedural. However, such skills are typically not cultivated with intentionality or systematically evaluated. To complement these partial findings, in this study we measure the self-perception of digital competence across three DigComp-aligned dimensions to examine whether esports participants report higher proficiency in these domains.

2.1.3. Job insecurity in the digital context

Job insecurity is a salient issue in the context of esports, particularly for participants who seek to professionalize their engagement. Unlike casual gamers, who approach gaming as a recreational activity

Table 1
Conceptual gaps in the literature on Esports participation.

Focus Area	Common Approach in the Literature	Gap Relevant to RQ1 and RQ2
Defining esports participants	Focus on competitive players; emphasis on psychological traits; no role differentiation	Overlooks non-playing, hybrid, and informal roles (→ RQ2); ignores subjective commitment (→ RQ2) and socio-demographic dimensions (→ RQ1)
Distinguishing esports participants vs. casual gamers	Differentiation based on external indicators (e.g., contracts, income), frequency of play, or competitive ranking	Fails to consider differences in digital skills, career values, and job insecurity perceptions (→ RQ1)

disconnected from economic expectations, esports participants may treat their involvement as a form of labor, which can be highly stressful [32]. Their activities may include streaming, competitive tournaments, and social media promotion, all of which lack contractual protection, stable income, or institutional support. Participants must be “prepared for sacrifice” [26], as job insecurity is not incidental but structurally embedded in the platform-mediated logic of digital labor [31,45]. This dynamic can be interpreted within broader transformations of work driven by digitalization and AI. On an optimistic view, AI reallocates routine, codifiable tasks to machines and augments human labor—raising productivity and freeing people for higher-value, non-routine activities [5]; however, a more critical literature documents risks as automation spreads such as displacement, routine jobs, and wage declines [1]. Sajja and Addula [42] state that lines between different jobs inside companies may begin to blur, which changes the organization’s structure. Situating esports within these transformations clarifies why boundaryless, hybrid role portfolios and perceived instability are not niche features, but reflect macro labor-market dynamics.

In Italy, these dynamics are particularly pronounced: studies show that players and organizers operate within a fragmented and weakly regulated ecosystem, characterized by informal arrangements and minimal prospects for long-term career sustainability [7,10]. In this study, we investigate whether self-identified esports participants report higher levels of perceived job insecurity than casual gamers, for whom gaming remains outside the domain of labor. Given the structural instability of esports labor, we expect that perceived job security will be significantly lower among esports participants—even among those with considerable skill or visibility.

2.2. Roles, Commitment, and career values in esports

Recent research [26,58] has argued that contemporary esports careers are built through hybrid practices that blur the lines between play and work, particularly as individuals shift between or combine roles such as competitor, streamer, organizer, or coach [7]. These hybrid arrangements partly reflect the absence of dedicated infrastructures and stable career paths, leaving participants reliant on logics that are familiar from platform-mediated gig work. Here, access to jobs, income generation and advancement are strongly tied to ratings and reputation [57] and, especially for creators/influencer-type roles, to visibility [52]. Such dynamics reflect a shift towards a market characterized by weaker protections for workers [29]. Considering the fragmentation typical of these digital environments, it becomes necessary to approach “esports participants,” not only as professional players, but also as ambiguous figures embedded in the broader online gaming ecosystem, where identities often overlap.

To better understand how individuals conceptualize their engagement in esports, we adopt Schein’s [44] framework of career anchors, a set of core values and self-perceived competencies that guide long-term career decisions. While originally developed in formal organizational settings, the concept has gained increasing relevance in informal and digitally mediated labor contexts, such as streaming, esports, and content creation [17]. In these fields, traditional markers of career advancements, such as hierarchical mobility or job stability, are often absent. As a result, individuals rely more heavily on intrinsic career values when deciding whether and how to pursue these paths. For example, those who prioritize autonomy, personal growth, or creative expression may find esports roles more compatible with their professional self-concept than conventional jobs. In contrast, individuals whose anchors center on job security or managerial advancement may be less inclined to pursue esports as a viable career path or may exit the ecosystem when structural instability becomes unsustainable. This distinction is reflected in recent research on digital labor. Törhönen et al. [50] found that streamers who aimed for professionalization treated their activity as a form of aspirational labor, investing time and resources in brand-building, audience engagement, and performance

work. Similarly, studies on esports participants show that motivations often revolve around mastery, recognition, and monetization [20,21,23], suggesting a strong alignment with anchors such as Technical Competence, Pure Challenge, and Entrepreneurial Creativity. By contrast, casual gamers tend to treat gaming as a recreational domain: for them, career identity is anchored in external domains and gaming remains a form of leisure. By applying the career anchor lens, our study explores how esports participation may reflect distinct career value orientations, and whether these differ systematically from those who do not engage in esports-related roles.

3. Method

This study aimed to investigate how Italian esports participants define their professional identity, differentiate themselves from those who take a casual approach to games, and perceive their career trajectories. To address these research questions, we conducted an online survey that combined ad hoc measures tailored to the esports context with validated scales adapted from prior research. The survey was designed to explore gaming habits, career values, digital competence, and job security perceptions, providing a structured approach to classifying different types of esports participants. The study was approved by the IRB of our institution. The full questionnaire and study materials are available on the Open Science Framework (OSF) repository,¹ allowing for full transparency and future replications of the study.

3.1. Sampling procedure

We employed a multi-channel recruitment strategy combining digital outreach, in-person engagement, and institutional endorsement to capture a diverse spectrum of esports participants. The first author contacted amateur and professional esports organizations via email and Instagram, as well as administrators of large Discord servers. An institutional endorsement from a recognized Italian esports organization (name anonymized) enhanced the study’s visibility and credibility. This allowed direct engagement with affiliated associations and networks. In-person recruitment occurred at a major gaming convention in [city anonymized], where the first author joined the event staff in the esports area for three days. Information was displayed on banners, and attendees were invited to complete the survey via QR codes, providing access to competitive players less likely to respond to online calls. We also leveraged social media, including sponsored Instagram posts and collaborations with streamers and esports organizations. Administrators of gaming communities on Facebook and Instagram promoted the study, as did an article in a widely-read online gaming magazine. Snowball sampling extended its reach: participants were encouraged to share the survey within their gaming networks. Physical outreach included distributing flyers with QR codes in gaming stores, universities, and public spaces. This ensured inclusion of participants not highly active in digital communities. A total of 515 individuals responded, including 99 self-identified as involved in esports-related activities (players, content creators, administrators, etc.). Of these, 388 completed the survey fully, including 74 esports participants. Eight cases were excluded due to missing consent. Eligibility criteria included being 18 or older and having gaming experience. The final sample comprised 297 complete and valid responses.

3.2. Participant characteristics

The final sample ($N = 297$) was composed largely of digitally literate, highly engaged young adults, predominantly male, who are familiar with online multiplayer environments and have a diverse range

¹ Please find the documents at the address: https://osf.io/axf2d/?view_only=d665cfd67bf3468d85ea1727ddf840b5.

of educational and occupational backgrounds. Table 2 provides further details on the characteristics of the sample.

3.3. Measures

To investigate the self-perceptions, career values, digital competence, and job-related concerns of Italian esports participants, we developed a structured online survey that combined ad hoc items designed for this study with validated instruments adapted from previous research. The survey was administered via [anonymized for peer review] platform, ensuring secure and anonymous participation. Prior to data collection, participants provided informed consent, in accordance with GDPR regulations and the ethical approval obtained from the [anonymized for peer review] Ethics Committee. The inclusion criteria required participants to be adults (18 +) residing in Italy and to have experience with video games, with a sub-section dedicated to those who declared themselves to be engaged in esports. Given that the survey was conducted in Italian, any scales not originally available in Italian were translated using a forward-backward translation procedure [33] to ensure conceptual and linguistic accuracy.² The translation process involved four translators, two of whom were native Italian speakers with professional expertise in translation and fluency in English, while the remaining two were native English speakers proficient in Italian. The first and second authors oversaw the entire process to ensure fidelity to the original meanings. In the first phase, the Italian native speakers independently translated the items from English to Italian, then compared their translations, resolving any discrepancies³ to create a unified version. In the second phase, the English native speakers, who had not seen the original version, translated this Italian version back into English. This back-translation was then compared to the original text, allowing the translators to address any inconsistencies and refine the final Italian version of the scales. After collecting demographic information (age, gender, education), the questionnaire was organized into four thematic areas. Table 3 summarizes the key measures used in the study, including their source, selected dimensions, and features.

Gaming Habits and Professional Involvement. This section of the survey includes a set of ad hoc questions designed to create a more precise and comprehensive taxonomy of esports participants. It begins by examining gaming habits, allowing for a distinction between casual participants and those with a structured, long-term commitment to esports. The sub-section dedicated to esports involvement not only differentiates competitive players from those in supporting roles (e.g., coaches, administrators, content creators) but also assesses their perceived level of professionalization, asking how individuals frame their own engagement with esports. A final sub-section explores the working conditions of individuals involved in esports, collecting data on job stability, employment sector, and working hours, to assess the economic sustainability of esports careers and the degree of overlap between competitive gaming and traditional employment pathways.

Digital Competence. The scale was adapted from the Job Digital Competence Scale (JDACS, [43], developed through a systematic review of DigComp-based assessment tools used in studies on working-age

² The finalized Italian questionnaire, alongside the process of translation, is available on this study's OSF repository to ensure transparency and reproducibility.

³ Some terms, such as "rotational assignment" and "functional skills," lacked direct Italian equivalents and required conceptual adaptation to ensure clarity. Similarly, idiomatic expressions, such as "I dream of being so good at what I do that my expert advice will be sought continuously," posed difficulties in capturing both the aspirational tone and the temporal dimension in a natural Italian form. Syntactic adjustments were necessary in cases where a literal translation resulted in awkward phrasing. Lastly, specific terminology related to career development and job security required careful refinement to align with Italian labor market concepts. For instance, "employment security" was translated to "sicurezza lavorativa" to reflect terminology more commonly used in Italian.

Table 2

Characteristics of the sample.

Socio-professional characteristics	
Gender	Male 81 %; Female 19 %
Age	M = 29.4 (SD = 7.16), MIN = 18, MAX = 62
Education	Middle school diploma: 18 %; High school diploma: 52 %; University degree: 25 %; Postgraduate degree: 6 %
Occupational status	Unemployed: 11 %; Student: 18 %; Employed: 53 %; Freelancer: 17 %; Retired individuals or homemakers: 1 %
Type of contract	Permanent: 71 %; Temporary: 18 %; Internship or other arrangements: 11 %
Working time regime	Full-time: 88 %; Part-time: 10 %; Project-based/On-Call: 2 %
Sector	Private: 75 %; Public: 25 %
Gaming habits	
Preferred gaming platform	PC: 50 %; PlayStation: 38 %; Xbox: 8 %; Other (Switch, mobile, retro): 4 %
Preferred gaming type	Singleplayer: 21 %; Local multiplayer: 6 %; Online multiplayer: 73 %
Continuity of past gaming	Mostly continuously: 72 %; With short breaks: 16 %; With long breaks: 12 %
Gaming approach	Only for fun: 28 %; Both for fun and competitively: 59 %; Mostly competitively: 10 %; Only competitively: 3 %
Weekly gaming frequency	< 1 day/week: 4 %; 1 day/week: 2 %; 2/3 days/week: 13 %; 4/5 days/week: 38 %; Every day: 43 %
Average gaming session duration	≤ 1 h: 3 %; 1–2 h: 17 %; 2–3 h: 33 %; 3–4 h: 29 %; 4–5 h: 10 %; > 5 h: 8 %

Table 3

Overview of the survey measures.

Measure	Source	Dimensions	Type of instrument
Gaming habits and professional involvement	Ad hoc	Gaming frequency, platform, esports role, perceived commitment, employment status	Custom, originally in Italian
Digital Competence	JDACS [43]	Communication & Collaboration; Digital Content Creation; Problem Solving	Adapted, originally in Italian
Career Anchors	Schein [44], Lazarova et al. [30]	Nine career anchors	Adapted, translated
Job Insecurity	[35]	Job Loss, Job Changes	Adapted, translated

adults, interviews with experts, and judge evaluations. For this study, we checked both the final instrument and the 61-item pool that was administered to judges. We selected and reformulated the items that were deemed particularly relevant to esports players, resulting in a 22-item instrument (7-point Likert-scale) across the five dimensions of DigComp 2.2: Information and data literacy, Communication and collaboration, Digital content creation, Safety, Problem solving. For the purpose of this study, we employed only three dimensions, which were the most directly relevant to esports and gaming practices⁴: communication and collaboration (6 items, $\omega = 0.73$, example item "I know how to use online services to share digital content"), digital content creation (5 items, $\omega = 0.79$, example item "I know how to use digital tools to create multimedia content"), problem solving (5 items, $\omega = 0.89$, example item "I can troubleshoot the operating system of my devices

⁴ Items from JDACS areas such as Information and Data Literacy (e.g., cross-checking information across multiple sources; using data-analysis software for work decisions) and Safety (e.g., adjusting firewall settings; detecting phishing attempts) are crucial components of general digital competence, but remain tangential to our focus on gaming practices. By contrast, the dimensions of Collaboration, Content Creation, and Problem Solving better capture skill domains that align directly with the demands of gaming.

independently”).

Career Anchors. We translated and adapted 32 items from Schein’s Career Anchors Orientation Inventory [44], selecting those most relevant to esports career paths. Following Lazarova et al. [30], certain items were excluded a priori due to their ipsative nature or potential overlap with other career anchors. Additionally, we integrated 4 items from Lazarova et al. [30] on the Internationalism Career Anchor, considering that many esports careers involve international competitions, cross-border collaborations, and global mobility. The final instrument consisted of 36 items (7-point Likert-scale), four for each of the nine career anchors. Composite reliability was satisfactory for all anchors: Technical/Functional Competence (TF, $\omega = 0.64$); General Managerial Competence (GM, $\omega = 0.77$); Autonomy/independence (AU, $\omega = 0.80$); Security/Stability (SE, $\omega = 0.75$); Entrepreneurial Creativity (EC, $\omega = 0.88$); Service/Dedication to a cause (SV, $\omega = 0.82$); Pure Challenge (CH, $\omega = 0.89$); Lifestyle (LS; $\omega = 0.75$); Internationalisation (INT; $\omega = 0.87$).

Job Insecurity. We translated twelve items from the Multidimensional Job Insecurity Measure (JIM; [35], specifically six items from the Job Loss ($\omega = 0.89$) and six from the Job Changes dimensions ($\omega = 0.91$) which capture cognitive and emotional dimensions of job insecurity, including concerns about job loss, perceived job instability and decline in working conditions, and anticipated resource constraints affecting job performance. Items related to purely organizational dimensions were instead excluded, as they may not be universally applicable to all participants. Participants answered on a 7-point Likert-scale. An example item for Job Loss was “I am certain of losing my job”, while an example item for Job Changes was “I will probably lose many features of my job that I value the most”. This scale was administered exclusively to participants who reported being currently employed, through conditional logic embedded in the survey that filtered out this section for people who declared not to be in the labor force.

3.4. Data analysis

Given the exploratory aim and the limited number of esports participants retained for subgroup analysis (74 out of 99), we adopted a mixed descriptive–comparative approach. Multivariate and latent models were excluded due to small subgroup sizes, potential model instability, and poor class separation. Rather than producing a definitive typology, we aimed to identify meaningful patterns in how esports participants conceptualize roles and occupational orientations within a fragmented ecosystem. Analyses were conducted in Jamovi (Version 2.3.21). We computed descriptive statistics for the full sample ($N = 297$) and the two subgroups: casual gamers and esports participants. Internal consistency of multi-item scales was assessed via McDonald’s Omega. We checked skewness, kurtosis, and ran Shapiro-Wilk tests, all significant at $p < 0.001$, indicating non-normality. For RQ1, we examined group-level differences in sociodemographics, gaming habits, digital competence, career values, and job insecurity using chi-square tests (nominal variables) and Mann-Whitney U tests (ordinal/continuous variables). Effect sizes were reported via Cramér’s V and rank-biserial correlation [13]. To address RQ2, we analyzed intra-group variation among esports participants. Descriptive and chi-square analyses were used to examine how functional roles (e.g., player, content creator) and perceived commitment levels (hobby, secondary, main occupation) related to gaming frequency, digital competence, career anchors, and job insecurity. As participants could report multiple roles, we generated a derived variable with five functional categories. Commitment was collapsed into three levels. Inferential results were interpreted with reference to theoretical frameworks in Sections 2 and 5. Multinomial logistic regression was excluded due to sample constraints. Models with < 10 events per variable (EVP) produce biased estimates and poor convergence [37]. Even with EPVm = 10, small samples can impair model performance [16]. In our case, modeling five role types against three commitment levels would require ≥ 80 participants per outcome

category, exceeding the esports sample. Chi-square tests were thus preferred for their lower distributional demands.

4. Results

This section provides an overview of the key findings from our study, structured around the two main research questions (refer to Table 4 for a summary). First, to provide a profile of esports participants (RQ1), we compared them with casual gamers across a range of relevant variables. Significant differences were found in education, gaming habits, and digital communication skills. Esports participants tended to play more competitively, preferred PC gaming, and reported greater engagement with online multiplayer games. However, no significant group differences were observed in employment status, job insecurity, or career anchor profiles. Second, we examined how esports participants perceive their activity (RQ2). Most did not consider esports a full-time profession, and only a small fraction viewed it as their main or economically viable occupation. Time investment alone does not appear to determine levels of commitment, although this relationship was not formally tested. Instead, functional roles showed a strong association with professionalization: participants involved in administrative roles were more likely to consider esports a primary occupation, whereas those identifying solely as players tended to describe their engagement as a hobby.

4.1. Profiling esports participants in comparison to casual gamers (RQ1)

In this section, we answer the first research question, namely, what are the socio-demographic and professional characteristics, gaming habits, perceived digital competence, career anchors, and job insecurity perceptions of those who identify as esports participants compared to those who do not? If differences exist, are they statistically significant? We examined this set of characteristics, comparing individuals who identified as involved in esports with those who did not, and found that the two groups were significantly different with respect to education, preferred gaming platform, preferred game types, gaming approach, weekly gaming frequency, average gaming session duration, digital communication and collaboration competence. Overall, effect sizes ranged from small to large. The gaming approach showed the strongest association with esports participation ($V = 0.35$, large effect). Education, preferred gaming platform, average gaming session duration, and weekly gaming frequency all had moderate effects ($V = 0.20\text{--}0.23$). Finally, the Mann-Whitney U test for digital communication and collaboration skills showed a small effect ($r = 0.19$). Table 5 represents an overview of this analysis.

4.1.1. Sociodemographic characteristics

The sample consisted of 297 participants, primarily digitally literate and highly engaged male young adults who were familiar with online multiplayer environments. Esports participants represented 25 % of the sample ($n = 74$), having self-identified as such, while the remainder

Table 4
Key takeaways from the results.

Theme/ RQ	Key takeaways
4.1 (RQ1)	● Esports participants are slightly younger with lower levels of higher education compared to non-esports participants. ● Esports participants prefer PC gaming and online multiplayer games. Both groups play frequently with long gaming sessions. ● Esports participants report significantly higher digital communication and collaboration skills.
4.2 (RQ2)	● Most esports participants do not consider esports a full-time job, but rather a secondary occupation or serious hobby. ● Among those who consider esports a job, the vast majority report insufficient or unsustainable earnings ● Functional roles matter. Players are least likely to view esports as a job, differently from administrators.

were classified as casual gamers. Gender distribution was nearly identical across the two groups, with a majority identifying as male. Although the median age of esports participants was slightly lower than that of casual gamers, this difference was not statistically significant. In contrast, significant group differences emerged in terms of educational attainment: esports participants were more likely to have completed only middle school, whereas casual gamers more frequently reported holding a university degree or postgraduate qualification.

4.1.2. Gaming preferences and patterns of engagement

Significant differences were observed in gaming platforms, genre preferences, and intensity of engagement. Esports participants showed a clear preference for PC gaming compared to casual gamers, while the latter group was more likely to favor PlayStation. Regarding game type, esports participants predominantly preferred online multiplayer games (with the majority of esports participants being affiliated with a single franchise—Call of Duty, as shown in Table 6). In terms of engagement, esports participants were more likely to play daily and reported longer gaming sessions, but frequent and extended gameplay is not exclusive to esports participants alone, since casual gamers also reported high-intensity engagement. Moreover, no significant group differences were observed regarding continuity of gaming over time (e.g., consistent vs. intermittent engagement). The two groups were more clearly differentiated in terms of their gaming approach, since esports participants were more likely to describe their activity as primarily competitive, while many casual gamers played mainly for enjoyment or a mix of motivations and only a minority identified as competitive players.

4.1.3. Digital skills

Perceived digital competence was assessed across three DigComp-aligned dimensions. A statistically significant difference was found only for digital communication and collaboration, where esports participants scored higher than casual gamers. No significant differences emerged in digital content creation or problem solving.

4.1.4. Career anchors

Career anchor profiles revealed no significant group-level distinctions across the nine anchor types. While descriptive trends suggested that esports participants may slightly prioritize managerial (GM) and entrepreneurial (EC) anchors more than casual gamers, and casual gamers tend to score marginally higher on security (SE) and lifestyle (LS), these variations were not statistically meaningful.

4.1.5. Job insecurity

No statistically significant differences were observed in occupational status or job insecurity perceptions, with both groups reporting low to moderate levels of perceived insecurity. However, in descriptive terms, esports participants were more likely to be employed and to hold permanent contracts, while casual gamers included a higher proportion of students and unemployed individuals. Casual gamers were also more likely to work full-time, potentially suggesting different patterns of workload or time management between the groups.

4.2. Esports Commitment, roles and professionalization (RQ2)

In this section, we address the second research question (RQ2), which investigates how esports participants differ in terms of the roles they occupy, the degree of professional commitment they express, and their orientation toward esports as a viable career path. To explore this question, we analyzed the distribution of functional roles, self-reported levels of professional commitment, esports seniority, and perceived economic sustainability. We also examined how functional role configurations are associated with different levels of professional engagement. Tables 6 and 7 present an overview of esports-related characteristics within this subgroup.

Table 5

Features of participants involved in Esports and casual gamers.

	Participation in Esports		p	Statistic	Effect size (V/r)
	No (n = 223)	Yes (n = 74)			
	%	%			
Gender			1.00	$\chi^2 (1, n = 296) = 0.00$	
Male	80.7 %	81.1 %			
Female	19.3 %	18.9 %			
Education			0.002	$\chi^2 (3, n = 297) = 15.05$	V = 0.23
Middle school diploma	14.3 %	28.4 %			
High-school diploma	50.2 %	55.4 %			
University degree	27.8 %	16.2 %			
Postgraduate degree	7.7 %	0.0 %			
Preferred gaming platform			0.006	$\chi^2 (5, n = 297) = 16.18$	V = 0.23
PC	43.9 %	68.9 %			
PlayStation	42.6 %	24.3 %			
Xbox	8.7 %	4.1 %			
Nintendo Switch	2.7 %	0.0 %			
Mobile	1.3 %	1.4 %			
Retro consoles	0.4 %	1.4 %			
Preferred game type			0.008	$\chi^2 (2, n = 297) = 9.74$	V = 0.18
Singleplayer	25.1 %	8.1 %			
Local multiplayer	5.8 %	6.8 %			
Online multiplayer	69.1 %	85.1 %			
Continuity of gaming			0.473	$\chi^2 (2, n = 297) = 1.50$	
Mostly continuously	70.4 %	77.0 %			
With short breaks	17.9 %	12.2 %			
With long breaks	11.7 %	10.8 %			
Gaming approach			0<.001	$\chi^2 (3, n = 297) = 37.42$	V = 0.35
Only for fun	33.2 %	12.2 %			
Both for fun and competitively	59.6 %	56.8 %			
Mostly competitively	6.7 %	21.6 %			
Only competitively	0.4 %	9.5 %			
Weekly gaming frequency			0.019	$\chi^2 (4, n = 297) = 11.74$	V = 0.20
< 1 day/week	4.9 %	0.0 %			
1 day/week	2.7 %	1.4 %			
2–3 days/week	15.2 %	5.4 %			
4–5 days/week	38.1 %	37.8 %			
Every day	39.0 %	55.4 %			
Average gaming session duration			0.009	$\chi^2 (5, n = 297) = 15.26$	V = 0.23
≤ 1 h	3.6 %	1.4 %			
1–2 h	20.6 %	8.1 %			
2–3 h	35.4 %	27.0 %			
3–4 h	25.1 %	39.2 %			
4–5 h	9.4 %	10.8 %			
> 5 h	5.8 %	13.5 %			
Occupational status			0.283	$\chi^2 (5, n = 292) = 6.24$	
Unemployed	13.5 %	4.3 %			
Student	18.0 %	18.8 %			
Employed	50.2 %	62.3 %			
Freelancer	17.5 %	14.5 %			
Retired	0.4 %	0.0 %			
Homemaker	0.4 %	0.0 %			

(continued on next page)

Table 5 (continued)

	Participation in Esports		<i>p</i>	Statistic	Effect size (<i>V/r</i>)
	No (<i>n</i> = 223)	Yes (<i>n</i> = 74)			
	%	%			
Type of contract			0.710	χ^2 (3, <i>n</i> = 155) = 1.38	
Permanent	69.6 %	74.4 %			
Temporary	17.8 %	18.6 %			
Internship	7.1 %	2.3 %			
Other	5.4 %	4.7 %			
Working time			0.158	χ^2 (2, <i>n</i> = 155) = 3.98	
Full-time	90.2 %	83.7 %			
Part-time	7.1 %	16.3 %			
Project-based/On-call	2.7 %	0.0 %			
	Median (IQR)	Median (IQR)			
Age (yrs)	28 (8.00)	27 (6.00)	0.191	<i>U</i> = 7414.50	<i>r</i> = 0.19
DC – Communication and Collaboration	6.17 (1.17)	6.33 (0.96)	0.014	<i>U</i> = 6688.00	
DC – Digital Content Creation	5.20 (1.80)	5.20 (1.40)	0.415	<i>U</i> = 7730.00	
DC – Problem Solving	5.40 (2.0)	5.20 (1.80)	0.594	<i>U</i> = 7909.50	
TF	5.50 (1.50)	5.38 (1.38)	0.572	<i>U</i> = 7890.50	
GM	4.00 (2.25)	4.50 (1.50)	0.133	<i>U</i> = 7290.00	
AU	6.00 (1.75)	5.75 (1.25)	0.763	<i>U</i> = 8059.00	
SE	6.25 (1.25)	6.13 (1.75)	0.669	<i>U</i> = 7979.50	
EC	5.00 (2.50)	5.00 (2.19)	0.470	<i>U</i> = 7788.50	
SV	5.25 (2.00)	5.13 (1.25)	0.971	<i>U</i> = 8227.50	
CH	5.50 (2.00)	5.00 (2.00)	0.530	<i>U</i> = 7850.00	
LS	6.00 (1.25)	5.75 (1.50)	0.204	<i>U</i> = 7443.50	
INT	5.25 (2.00)	5.13 (1.25)	0.971	<i>U</i> = 8227.50	
Job insecurity	2.58 (1.75)	2.29 (1.85)	0.809	<i>U</i> = 4730.50	

Effect sizes reported only for significant differences.

4.2.1. Esports seniority and functional roles

While some participants were relatively new to the scene, the majority reported long-term involvement, with many having participated in esports for over five years. Participants reported a wide range of roles, reflecting the heterogeneity of the esports ecosystem. In addition to gameplay, many were engaged in content creation, streaming, or administrative and management activities. Although some individuals identified with a single role, a substantial portion described hybrid configurations. Four main types of functional roles emerged: pure players, content creators, administrators, and hybrid profiles combining either gameplay with content production or administrative tasks with other responsibilities. Details are reported in Table 6.

4.2.2. Association between commitment and functional roles

When asked to describe their professional status, most participants placed themselves between hobbyist and semi-professional positions; some viewed esports as a serious activity that demanded discipline and planning, but fell short of being considered employment; others described it as a complementary or transitional occupation, pursued

Table 6

Esports-related variables.

Variable	%
Esports seniority	18.9 % less than 1 year 12.2 % 1 year 17.6 % 1–2 years 10.8 % 2–3 years 17.6 % 3–4 years 23.0 % 5 years or more
Main game Role	81.1 % Call of Duty 38.6 % Only player 14.3 % Only content creation 20.0 % Player and content creator 17.1 % Only administration and management 10.0 % Administration plus non-admin roles
Esports commitment	14.9 % No professional implication 36.5 % Serious activity, but not considered a job 27.0 % Considered it a job, but not their main job 14.9 % Considered it a job, alongside other job activities 1.4 % Principal professional activity, but with a side hustle 5.4 % Only professional activity
Esports economic sustainability	88.6 % not economically sustainable

alongside other forms of labor, often with the hope of future professionalization; only a small subset reported treating esports as their primary professional activity. Table 6 reports the distribution of the different levels of commitment. A closer examination of the relationship between functional roles and perceived commitment revealed consistent patterns. Participants who identified solely as players were the most likely to describe their esports activity as non-professional. Within this group, the majority stated that their engagement had no professional implication, with only a small minority framing it as a job-related activity. By contrast, those involved in administrative functions, either exclusively or in combination with other roles, were more likely to report higher levels of professional commitment. Many described esports as a secondary job or as part of a broader portfolio of professional activities. Notably, participants with hybrid profiles, such as those who combined playing and content creation, also exhibited a higher degree of occupational identification compared to single-role participants. These individuals were more likely to see their esports engagement as a structured and purposeful endeavor, even when it was not their primary occupation. Table 7 summarizes this distribution.

4.2.3. Perceived commitment and economic sustainability

Although almost half of the participants expressed a strong commitment to their esports activity, whether as a job or as a structured long-term pursuit, most did not perceive it as economically sustainable. Nearly 90 % of participants of this subgroup stated that their esports engagement did not constitute a viable source of income; an unsurprising result, considering that most of the participants declared that they combined esports professional participation with other forms of work or study. Considering this polarizing result, we did not explore further associations with this variable, which likely reflects the field's under-institutionalization and limited monetization for most participants [10,40].

5. Discussion

Our findings reveal modest differences between esports participants and casual gamers, which mostly concern gaming practices. However, these differences do not consolidate into clearly distinct labor conditions or occupational identities. Instead, more meaningful differences emerge when focusing within the esports subsample, where functional roles are associated with perceived professional commitment to esports as a more or less work-oriented endeavor. These elements, taken together, invite a conceptual shift: from quantifying behavioral engagement to understanding how individuals perceive their engagement with esports depending on their role. The following sections interpret our findings

Table 7

Distribution of Esports commitment across different roles.

Esports commitment		Role Player	Content Creator and Streaming	Player and Content Creator	Administration and Management	Administration and Non- admin roles
Not a professional activity	% Row	63 %	9 %	20 %	0 %	9 %
	% Column	81 %	30 %	50 %	0 %	43 %
Secondary professional activity	% Row	16 %	26 %	16 %	26 %	16 %
	% Column	11 %	50 %	21 %	42 %	43 %
Professional activity	% Row	13 %	13 %	25 %	44 %	6 %
	% Column	7 %	20 %	29 %	58 %	14 %

through this lens, drawing on theoretical frameworks such as aspirational labor and boundaryless careers.

5.1. Continuities and discontinuities in esports and casual gaming engagement

Building on the quantitative analyses presented in Section 4.1, this section discusses the socio-demographic and behavioral similarities and distinctions between esports participants and casual gamers. Our findings reveal that while these two groups differ in certain practices and value orientations, they do not represent socio-professionally distinct categories. Instead, they occupy positions along a continuum of engagement, where differences in age, skills, and commitment intersect without forming consolidated occupational identities.

5.1.1. Gender, age, and education: A profile of informal participation

From a socio-demographic standpoint, we found both common ground and subtle differences. Gender distributions were similarly skewed across both groups, with a predominance of male participants. While this pattern reflects broader demographic trends in competitive gaming, with a predominance of male participants [41], the presence of a non-negligible proportion of female respondents suggests that women may face fewer access barriers to esports participation than is commonly assumed. Esports participants also tended to be slightly younger and less formally educated than casual gamers, who more frequently reported university degrees while esports participants were more likely to hold only a middle or high school diploma. This gap may reflect the age distribution of esports participants—particularly among players—or their preference for early workforce entry due to limited economic prospects in the sector [26]. It is also plausible that esports participants prioritize experiential digital expertise over formal academic credentials.

5.1.2. Gaming practices and digital competence

Differences between the two groups concern three main behavioral domains: (a) platform preferences, (b) digital collaboration skills, and (c) intensity of play. Esports participants tend to favor PC-based games, particularly those supporting online multiplayer formats. This preference likely reflects the competitive structure of esports, where PCs offer high performance, peripheral compatibility, and customizability; all crucial elements in tournament settings. Moreover, LAN-based tournaments remain standard practice within esports [56] to ensure fairness [46]. Esports participants also reported slightly higher digital collaboration and communication skills, one of the core areas of the DigComp 2.2 framework [54] that reflects the technical demands of hybrid esports roles. These findings align with studies emphasizing the informal acquisition of these skills through gameplay, such as teamwork [34,59], team coordination and leadership [22]. However, since no differences emerged in the other DigComp dimensions measured (e.g., content creation, problem-solving), these skills may not be uniformly distributed across all esports participants but concentrated in those occupying specific functional roles (a hypothesis developed in Section 5.2).

Interpreting these patterns through the lens of digital self-efficacy [18], DSE may serve as an enabling mechanism for competence enactment. However, to date, there is no evidence that esports participants score higher on general DSE than comparable peers; existing work either examines gameplay-specific efficacy [55] or general self-efficacy within esports samples [11]. Evidence from outside esports (e.g., [2] cautions against assuming a linear relationship between DSE and engagement, stating that the adoption of a technology is context-contingent and structural factors (such as costs) weigh in. For esports, these context-contingent effects support measuring role-specific DSE (e.g., confidence in live streaming, event coordination) in esports samples, alongside DigComp, to test its association with perceived professional commitment and the framing of gaming as work.

In terms of engagement with play, esports participants reported slightly more frequent and intense activity, aligning with the training demands of competitive play [39]. Yet casual gamers also demonstrated high levels of continuity, frequency, and session length. This has two key implications: (1) time investment alone may not be a reliable proxy for professional engagement; and (2) casual gamers may adopt similar behavioral patterns to esports participants, blurring the line between leisure and pre-professional activity. These findings support the need for a more nuanced analytical framework that goes beyond behavioral metrics to consider structural and identity-based factors.

5.1.3. Career anchors and occupational boundaries

Turning to professional values, esports participants tended to score higher on career anchors associated with autonomy and entrepreneurial creativity, although these differences were not statistically significant. This descriptive trend may indicate an emerging orientation toward self-directed or innovative career paths, even if the distinction from casual gamers is not clearly delineated. This is consistent with research on aspirational labor and informal digital economies [17], where individuals often pursue professional ambitions—such as aspiring to become full-time esports players or streamers—before formal structures like contracts or institutional recognition are in place. One possible explanation for the lack of statistical differentiation lies in the precarious and under-institutionalized nature of the Italian esports sector, where clear career pathways are scarce and occupational identities are still in flux. Finally, when examining structural employment variables (contract type, job status, and perceived job insecurity) no significant group differences emerged. Despite extensive literature framing esports labor as precarious and unstable [45,48,58], our data suggest that such conditions are not explicitly perceived as insecurity by participants. This may be due to the fact that esports engagement appears to remain complementary or marginal to individuals' core occupational identities and it is not fully framed as a "job." Taken together, these findings suggest that while behavioral and value-based distinctions exist, they have not yet been crystallized into clearly differentiated occupational roles or identities. This points to the persistence of informal, fluid engagement models in esports, rather than stable professional roles, as explored in the next section.

5.2. From gameplay to role-based commitment: Understanding esports engagement beyond time and rank

Building on the quantitative analyses presented in Section 4.2, this section dives into the category of esports participants. Our findings challenge the assumption that time investment or competitive performance are the main indicators of professional engagement in esports and highlight the importance of role specialization as a more meaningful predictor of perceived occupational identity. To discuss these results, we focus on core theoretical constructs, such as occupational identity, aspirational labor, and boundaryless careers.

5.2.1. Role-based commitment: Esports engagement beyond time and rank

Among those reporting a clear sense of professional commitment, the majority held multi-functional or managerial roles, while only a minority were players with no additional responsibilities. This means that individuals involved in multi-functional or *hybrid* roles (such as content creation, administration, team coordination) are more likely to describe their involvement as “work”. However, these roles, often combined with gameplay, may reflect a form of informal professionalization rooted in what individuals do, rather than how frequently they play or how much they earn.

Moreover, life-course dynamics may explain such patterns. “Player-only” participants in our sample were on average younger and showed greater age dispersion, often still engaged in education or early-stage career trajectories: their lower levels of professional self-identification may thus also reflect a developmental phase in which occupational identity is still fluid. By contrast, participants in hybrid or managerial roles, who are typically older and more stable in their educational or employment status, reported stronger occupational commitment. This trend mirrors broader industry trends, where top-tier professional players often supplement competitive play with content creation, coaching, or organizational responsibilities as a strategy for visibility and economic sustainability [49]. This pattern aligns with observed dynamics in platform-mediated work, where reputation and audience visibility are central to stabilizing opportunities [17,52,57]. A similar convergence of functions supports the emergence of *blended professional identities* in the esports ecosystem, where individuals simultaneously occupy the roles of player, creator, and organizer. This phenomenon is increasingly common within the industry, as professional esports athletes transition into roles that combine competitive play with content creation and organizational responsibilities [14].

These hybrid careers may be particularly prominent in under-institutionalized contexts and platformized labor markets, characterized by weaker protections and fewer stable career ladders [29], and may be further amplified in settings such as Italy, where formal career pathways in the esports sector are scarce and professional boundaries remain porous [40]. In this context, hybridization is not accidental but structurally induced, reflecting a pragmatic response to weak institutional support and economic uncertainty—features typical of informal creative economies where stable, traditional career paths are replaced by “boundaryless careers”, which are self-directed trajectories that cross traditional professional and organizational boundaries [4]. The observed association between hybrid role-holding and stronger occupational identification is also consistent with a platform-oriented career logic: by combining roles that generate visibility and diversify opportunities (e.g., content creation, coordination, administration), participants show great commitment, therefore they are more likely to frame their engagement as work even under uncertainty.

In sum, functional role specialization acts as an informal mechanism of professionalization, shaping not only how participants engage with esports, but how they define themselves in relation to it. The next section explores how these roles combine into hybrid identities across the esports value chain.

5.2.2. A typology of esports engagement

To synthesize our findings and deepen the theoretical contribution, we propose a typology of esports engagement grounded in functional roles and self-perceived commitment. This heuristic, empirically grounded framework integrates functional roles, self-reported commitment, and perceived professionalization. 1) Player-hobbyist: Primarily engaged in gameplay for personal enjoyment, with no aspirations for monetization or professional status. These participants usually report high playtime but do not combine it with content production or team-based roles. Their self-identification remains ludic, and they explicitly reject the framing of esports as work; 2) Aspiring professional: Characterized by a future-oriented commitment to esports as a possible career path. These participants often compete seriously and invest in self-promotion, audience-building, or team integration, though they lack stable income. Their narratives reflect a symbolic orientation toward professionalism despite limited material returns. 3) Hybrid creator: Combines competitive play with platform-based roles such as streaming, content creation, or coaching. Often describes esports as semi-professional labor, dependent on visibility, engagement metrics, and reputation. Income is fragmented, and identity is built through audience interaction and multi-platform activity. 4) Administrator: Focuses on backstage roles such as event coordination, team management, or communication. These participants report the highest levels of professional commitment and task responsibility. While gameplay may be secondary, they view esports as work, despite limited institutional recognition or financial viability.

6. Limitations

This study has several limitations that help contextualize its findings. First, the recruitment strategy—based on digital networks and social media—may have introduced self-selection bias. The esports subgroup included a high proportion of Call of Duty players, likely due to genre-specific community structures and platform visibility. This overrepresentation may limit the generalizability of results to other segments of the esports population. Second, participants were classified based on self-perceived roles and professional commitment. While this aligns with bottom-up approaches in digital labor research, it does not capture external markers of professionalization. Future studies should consider triangulating self-reports with objective indicators of esports status. Third, the small esports sample limited the use of multivariate models, which are better suited to larger samples. Future research could employ larger, stratified samples to validate role typologies and track how identities and aspirations evolve over time. Finally, the study is context-specific: Italy’s under-institutionalized esports ecosystem may not reflect dynamics in more mature markets. Cross-national comparisons are needed to assess the transferability of these findings to different regulatory and cultural settings.

7. Conclusion

This study examined how esports participants in Italy construct their professional identities, roles, and aspirations within a precarious and informal labor market. While they differ from casual gamers in platform preference, collaboration skills, and competitive orientation, no major distinctions emerged in job status, perceived insecurity, or career values—suggesting that esports has not yet consolidated as a distinct socio-professional category. Crucially, our findings indicate that perceived professional commitment is more closely associated with functional role specialization than with time investment or performance level. Participants engaged in content creation, coordination, or administrative tasks expressed stronger occupational identification than pure players, supporting theoretical frameworks on hybrid, boundaryless, and aspirational careers, and situating these role-based skill configurations within gig-economy forms of work. To capture this diversity, we proposed a role-based typology that reflects the fluid and ambiguous nature of

esports labor. Specifically, the typology presented can inform the design of role-sensitive interventions (such as tailored training for hybrid creators, legal recognition for administrative roles, or platform support for aspiring professionals), ensuring that support policies align with the lived experiences of esports participants. Finally, by situating esports within a structurally weak and under-regulated context like Italy, this study highlights the sector's relevance as a site of emergent digital labor—one that requires greater scholarly, educational, and policy attention.

CRedit authorship contribution statement

Arianna Boldi: Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Domenico Sanseverino:** Writing – review & editing, Methodology, Formal analysis, Data curation, Conceptualization.

Ethics approval

This study was approved by the Ethics Committee of the University of Turin, Department of Psychology (Protocol n. 207981).

Consent to participate

Participation was anonymous, and data were collected through an online survey with informed consent embedded at the beginning of the questionnaire.

Consent for publication

Not applicable. The study does not include any identifiable personal data or images requiring publication consent.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data and study materials are available on the Open Science Framework (OSF) page of the project: https://osf.io/axf2d/?view_only=d665cfd67bf3468d85ea1727ddf840b5

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